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ME-CSE
Group-2

PCS223 Advanced Computer Networks

1. To capture and analyze ICMP packets (Ping request and reply)

icmp

No.	Time	Source	Destination	Protocol	Length	Info
342	2.253957	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12751/53041, ttl=1…
345	2.272124	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12751/53041, ttl=1…
511	3.277336	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12752/53297, ttl=1…
518	3.296305	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12752/53297, ttl=1…
659	4.295383	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12753/53553, ttl=1…
662	4.311457	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12753/53553, ttl=1…
806	5.309090	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12754/53809, ttl=1…
807	5.334587	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12754/53809, ttl=1…

```
▾ Internet Protocol Version 4, Src: 172.16.53.25, Dst: 142.250.182.238
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 60
    Identification: 0xa5ee (42478)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x6dc0 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.53.25
    Destination Address: 142.250.182.238
```

```
▾ Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x1b8c [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 12751 (0x31cf)
    Sequence Number (LE): 53041 (0xcf31)
    [Response frame: 345]
  ▸ Data (32 bytes)
```

2. To analyze TCP packets and observe the TCP three-way handshake

832	3.698988	172.16.53.25	142.251.43.174	TCP	66 57327 → 443 [SYN] Seq=0 Win=65535 L…
834	3.703986	142.251.43.174	172.16.53.25	TCP	66 443 → 57327 [SYN, ACK] Seq=0 Ack=1 …
835	3.704181	172.16.53.25	142.251.43.174	TCP	54 57327 → 443 [ACK] Seq=1 Ack=1 Win=6…
836	3.704888	172.16.53.25	142.251.43.174	TCP	1514 57327 → 443 [ACK] Seq=1 Ack=1 Win=6…
837	3.704888	172.16.53.25	142.251.43.174	TLSv1.3	394 Client Hello (SNI=encrypted-tbn0.gs…

```
Transmission Control Protocol, Src Port: 57327, Dst Port: 443, Seq: 0, Len: 0
    Source Port: 57327
    Destination Port: 443
    [Stream index: 7]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 2520892342
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x002 (SYN)
    Window: 65535
    [Calculated window size: 65535]
    Checksum: 0x979a [unverified]
```

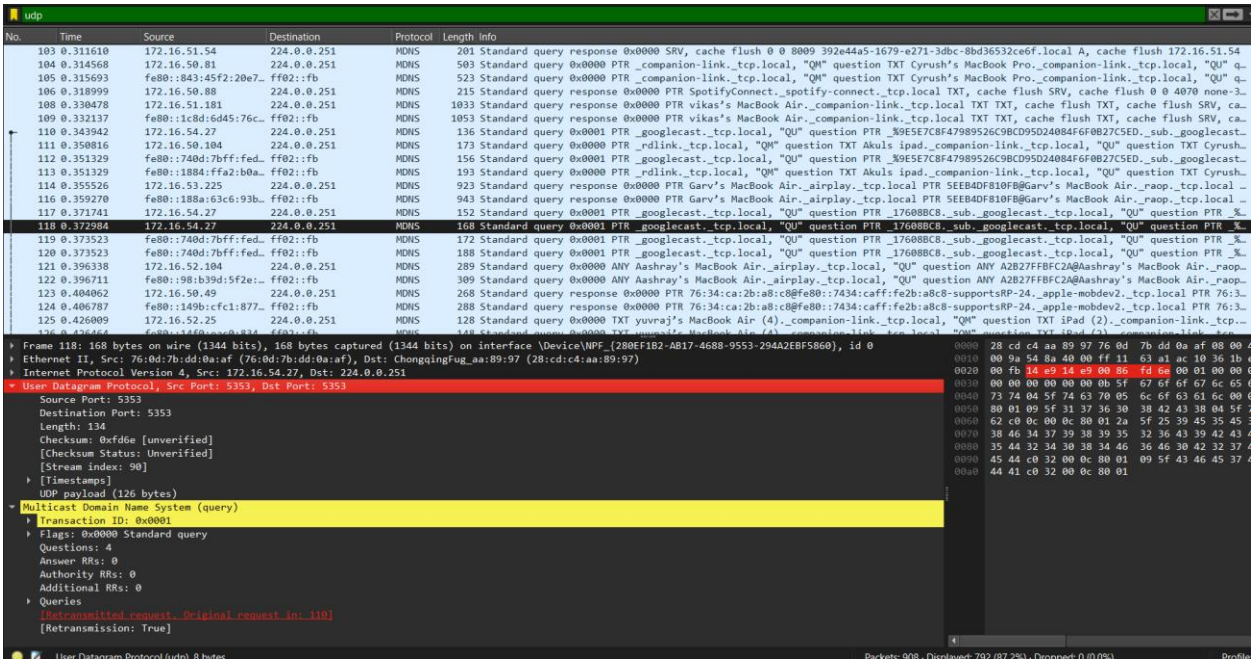
834	3.703986	142.251.43.174	172.16.53.25	TCP	66 443 → 57327 [SYN, ACK] Se
835	3.704181	172.16.53.25	142.251.43.174	TCP	54 57327 → 443 [ACK] Seq=1 A
836	3.704888	172.16.53.25	142.251.43.174	TCP	1514 57327 → 443 [ACK] Seq=1 A
837	3.704888	172.16.53.25	142.251.43.174	TLSv1.3	394 Client Hello (SNI=encrypt
844	3.735631	172.16.53.25	142.251.43.174	TCP	1514 [TCP Retransmission] 5732
845	3.739748	142.251.43.174	172.16.53.25	TCP	60 443 → 57327 [ACK] Seq=1 A
846	3.739748	142.251.43.174	172.16.53.25	TCP	60 443 → 57327 [ACK] Seq=1 A
851	3.747809	142.251.43.174	172.16.53.25	TCP	66 [TCP Dup ACK 846#1] 443 →

```
▸ Frame 834: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \De
▸ Ethernet II, Src: Fortinet_6e:91:19 (04:d5:90:6e:91:19), Dst: ChongqingFug_aa:89:97 (
▸ Internet Protocol Version 4, Src: 142.251.43.174, Dst: 172.16.53.25
▾ Transmission Control Protocol, Src Port: 443, Dst Port: 57327, Seq: 0, Ack: 1, Len: 0
    Source Port: 443
    Destination Port: 57327
    [Stream index: 7]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 592384709
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2520892343
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x012 (SYN, ACK)
    Window: 14600
    [Calculated window size: 14600]
    Checksum: 0x286d [unverified]
    [Checksum Status: Unverified]

0000  28 cd c
0010  00 34 c
0020  35 19 0
0030  39 08 2
0040  03 08
```

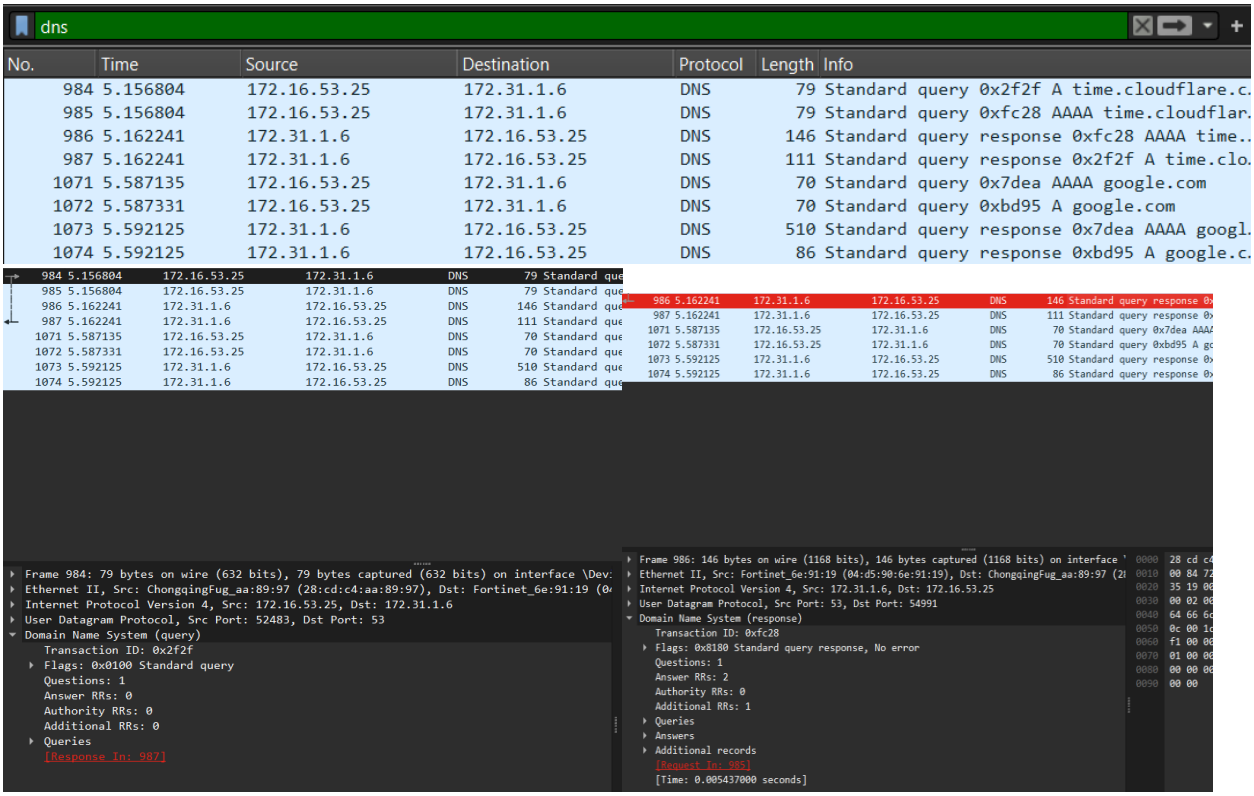
835	3.704181	172.16.53.25	142.251.43.174	TCP	54 57327 → 443 [ACK] Seq=1 Ack
836	3.704888	172.16.53.25	142.251.43.174	TCP	1514 57327 → 443 [ACK] Seq=1 Ac
837	3.704888	172.16.53.25	142.251.43.174	TLSv1.3	394 Client Hello (SNI=encrypte
844	3.735631	172.16.53.25	142.251.43.174	TCP	1514 [TCP Retransmission] 57327
845	3.739748	142.251.43.174	172.16.53.25	TCP	60 443 → 57327 [ACK] Seq=1 Ac
846	3.739748	142.251.43.174	172.16.53.25	TCP	60 443 → 57327 [ACK] Seq=1 Ac
851	3.747809	142.251.43.174	172.16.53.25	TCP	66 [TCP Dup ACK 846#1] 443 →

```
▸ Frame 835: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \De
▸ Ethernet II, Src: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97), Dst: Fortinet_6e:91:19 (
▸ Internet Protocol Version 4, Src: 172.16.53.25, Dst: 142.251.43.174
▾ Transmission Control Protocol, Src Port: 57327, Dst Port: 443, Seq: 1, Ack: 1, Len: 0
    Source Port: 57327
    Destination Port: 443
    [Stream index: 7]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2520892343
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 592384710
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x010 (ACK)
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    [Checksum: 0x149a [unverified]

0000  04 d5 90
0010  00 28 fe
0020  2b ae df
0030  00 ff a1
```

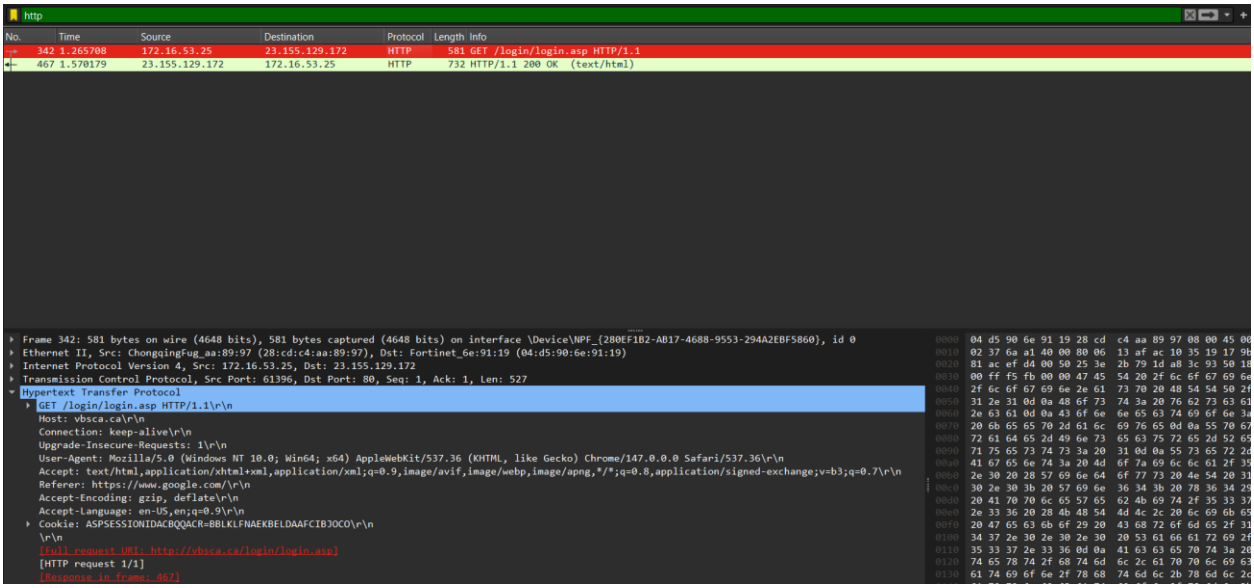
3. To capture and analyze UDP packets

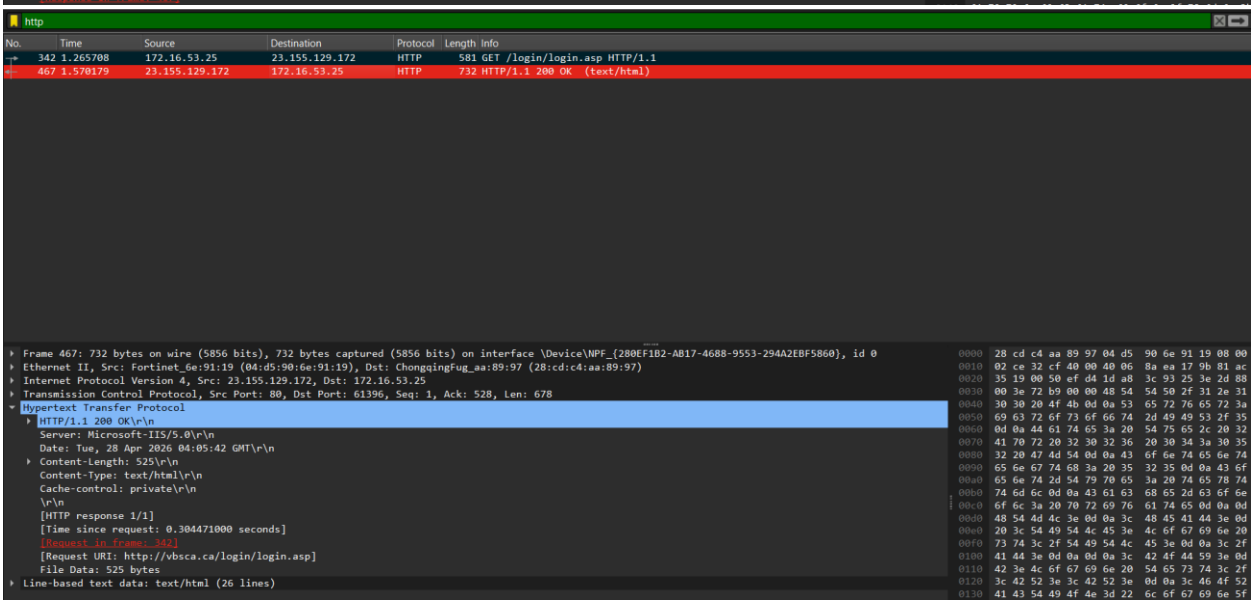


4. To capture and analyze DNS query and response packets



5. To capture and analyze HTTP packets





6. To analyze ARP request and reply packets

arp

No.	Time	Source	Destination	Protocol	Length	Info
91	1.532646	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
96	2.292872	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
102	3.282048	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
103	4.298448	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
104	5.286776	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
107	6.290745	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
112	7.298439	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
134	8.283832	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
136	9.289302	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
142	10.298956	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
146	11.293560	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
147	12.285999	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131

```
▸ Frame 91: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▸ Ethernet II, Src: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
    Sender IP address: 10.50.73.131
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.50.73.130
```

7. To identify source and destination MAC and IP addresses in captured Packets

317	2.256984	148.113.16.46	10.50.73.131	TCP	66	443 → 50262 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
318	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
319	2.256984	142.251.157.119	10.50.73.131	UDP	137	443 → 61979 Len=95
320	2.256984	64:ff9b::8efb:dd62	2409:40d1:0:9dc3:4d…	QUIC	87	Protected Payload (KP0)
321	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
322	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
323	2.257241	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
324	2.258892	10.50.73.131	142.251.157.119	UDP	86	61979 → 443 Len=44
325	2.259032	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
326	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
327	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
328	2.271864	142.251.157.119	10.50.73.131	UDP	309	443 → 61979 Len=267

```
▸ Frame 317: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▾ Ethernet II, Src: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4), Dst: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Destination: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Source: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4)
    Type: IPv4 (0x0800)
▸ Internet Protocol Version 4, Src: 148.113.16.46, Dst: 10.50.73.131
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 50262, Seq: 1, Ack: 2, Len: 0
```


317	2.256984	148.113.16.46	10.50.73.131	TCP	66	443 → 50262 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
318	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
319	2.256984	142.251.157.119	10.50.73.131	UDP	137	443 → 61979 Len=95
320	2.256984	64:ff9b::8efb:dd62	2409:40d1:0:9dc3:4d…	QUIC	87	Protected Payload (KP0)
321	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
322	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
323	2.257241	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
324	2.258892	10.50.73.131	142.251.157.119	UDP	86	61979 → 443 Len=44
325	2.259032	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
326	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
327	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
328	2.271864	142.251.157.119	10.50.73.131	UDP	309	443 → 61979 Len=267

```
▸ Frame 317: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▾ Ethernet II, Src: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4), Dst: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Destination: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Source: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4)
    Type: IPv4 (0x0800)
```


```
▾ Internet Protocol Version 4, Src: 148.113.16.46, Dst: 10.50.73.131
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0x0000 (0)
  ▸ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 43
    Protocol: TCP (6)
    Header Checksum: 0x5770 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 148.113.16.46
    Destination Address: 10.50.73.131
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 50262, Seq: 1, Ack: 2, Len: 0
```

```
0000  28 cd
0010  00 34
0020  49 83
0030  01 f5
0040  78 4a
```